

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Artificial Intelligence (AI) and Data Science) SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Artificial Intelligence (AI) and Data Science) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTAIC501 - COMPUTER NETWORK AND CLOUD COMPUTING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12**
- 1 Which multiple access protocol is used in Ethernet?
a) Token passing b) FDMA c) TDMA d) CSMA/CD
 - 2 Which of the following best describes the primary use of computer networks?
a) Data redundancy b) Resource sharing
c) Signal amplification d) Error correction
 - 3 DNS primarily translates _____.
a) MAC address to IP b) Domain names to IP addresses
c) IP to MAC d) Port numbers to IP
 - 4 AWS, Azure, and Google Cloud are examples of _____ cloud.
a) Private b) Public c) Community d) Hybrid
 - 5 Which virtualization technique isolates OS instances?
a) Full virtualization b) Hardware virtualization
c) OS-level virtualization d) Para-virtualization
 - 6 The count-to-infinity problem occurs in which routing algorithm?
a) Link-state routing b) Flooding
c) Source routing d) Distance-vector routing
 - 7 Which cloud service supports real-time communication?
a) MaaS b) CaaS c) IaaS d) PaaS
 - 8 Virtualization is a key enabling technology for cloud computing.
a) True b) False
 - 9

List A	List B
1. HTTP	A. File transfer
2. FTP	B. Web access
3. SMTP	C. Email sending
4. DNS	D. Name resolution
a) 1-B, 2-A, 3-C, 4-D	b) 1-A, 2-B, 3-D, 4-C
c) 1-C, 2-D, 3-A, 4-B	d) 1-D, 2-C, 3-B, 4-A
 - 10 Which protocol suite is used on the Internet?
a) OSI b) NetBIOS c) TCP/IP d) ATM

- | 11 | List A | List B |
|----|---|--------------------------------|
| | 1. IEEE | A. Internet standards |
| | 2. ISO | B. Electrical standards |
| | 3. IETF | C. International standards |
| | 4. ITU-T | D. Telecommunication standards |
| | a) 1-B, 2-C, 3-A, 4-D | b) 1-C, 2-A, 3-D, 4-B |
| | c) 1-A, 2-B, 3-C, 4-D | d) 1-D, 2-C, 3-B, 4-A |
| 12 | CRC is an example of _____ error detection technique. | |
| | a) Parity check | b) Polynomial based |
| | c) Hamming code | d) Checksum |
- Q2. Solve the following.
- A) Differentiate between client-server and peer-to-peer networks with examples. 6
- B) Name the layers of the OSI reference model. Explain the functionality of the each layer. 6
- Q3. Solve the following.
- A) Name different multiple access protocols. Compare TDMA and FDMA multiple access protocols. 6
- B) List common error detection techniques. Explain the need for error detection and correction. 6
- Q4. Solve Any Two of the following.
- A) Define transport services in computer networks. Explain in detail major application-layer protocols. 6
- B) Explain the TCP header format in detail with a suitable diagram. 6
- C) Identify the functions of DNS. Explain the working of DNS name resolution. 6
- Q5. Solve Any Two of the following.
- A) Define cloud computing. Describe public, private, and hybrid clouds. 6
- B) List key components of a data center. Compare traditional data centers with cloud data centers. 6
- C) Explain the benefits and risks of cloud computing. Explain basic cloud security concepts. 6
- Q6. Solve Any Two of the following.
- A) Define virtualization. Explain CPU virtualization techniques. 6
- B) Name key features of Amazon EC2. Explain the differences between IaaS, PaaS, and SaaS. 6
- C) List types of virtualization. Compare hypervisor-based and container-based virtualization. 6

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